

Capstone Section 3 - MySQL

All commands below were run against a local MySQL 8.0.40 server (mysql -u root < mysql-queries.sql). Output is captured directly from the server, not simulated.

Create instructors table & insert records

```
CREATE TABLE IF NOT EXISTS instructors (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE,  
    department VARCHAR(100)  
);  
  
INSERT INTO instructors (name, email, department) VALUES  
    ('Dr. Mei Lin', 'mei.lin@lms.edu', 'Computer Science'),  
    ('Dr. Arun Kumar', 'arun.kumar@lms.edu', 'Data Science'),  
    ('Dr. Wei Zhang', 'wei.zhang@lms.edu', 'Artificial Intelligence');  
  
SELECT * FROM instructors;
```

Result:

```
+----+-----+-----+-----+  
| id | name          | email                | department          |  
+----+-----+-----+-----+  
|  1 | Dr. Mei Lin   | mei.lin@lms.edu     | Computer Science    |  
|  2 | Dr. Arun Kumar | arun.kumar@lms.edu | Data Science        |  
|  3 | Dr. Wei Zhang | wei.zhang@lms.edu  | Artificial Intelligence |  
+----+-----+-----+-----+
```

Add User + Enroll + JOIN Query

```
-- 1. Add a new user  
INSERT INTO users (name, email) VALUES ('Cara Ong', 'cara.ong@example.com');  
  
-- 2. Enroll that user in a course  
INSERT INTO enrollments (user_id, course_id)  
VALUES (  
    (SELECT id FROM users WHERE email = 'cara.ong@example.com'),  
    (SELECT id FROM courses WHERE title = 'AI & Machine Learning')  
);  
  
-- 3. JOIN query showing the enrolled user with their course and instructor  
SELECT  
    users.name          AS student_name,  
    courses.title        AS course_title,  
    instructors.name     AS instructor_name,  
    enrollments.enrolled_at  
FROM enrollments  
JOIN users      ON enrollments.user_id = users.id  
JOIN courses    ON enrollments.course_id = courses.id  
JOIN instructors ON courses.instructor_id = instructors.id;
```

Result:

student_name	course_title	instructor_name	enrolled_at
Cara Ong	AI & Machine Learning	Dr. Wei Zhang	2026-09-27 11:41:33

The instructors, users, courses and enrollments schema (with AUTO_INCREMENT primary keys, a UNIQUE constraint on instructors.email, and foreign keys linking enrollments back to users/courses) is in mysql-queries.sql.